

**MODEL 020C
WIND DIRECTION SENSOR
OPERATION MANUAL**



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1.0 GENERAL INFORMATION

- 1.1 The Model 020 Wind Direction Sensor uses a lightweight, vane and a microtorque potentiometer to produce an analog voltage output proportional to wind direction. An internal heater reduces moisture for extended bearing life. The sensor mounts in the 191 Crossarm Assembly.
- 1.2 The sensor cable has a quick-connect connector with vinyl-jacketed, shielded cable. Cable length is given in -XX feet on each cable part number. A 2108-XX cable is supplied for translators having circular type cable connectors, and a 1957-XX cable is supplied for translators having terminal strip connections. (Color code, FIGURE 2-1.)

Table 1-1
Model 020C Wind Direction Sensor Specifications

Performance

Azimuth	Electrical 0-356° Mechanical 0-360°
Threshold	0.6 mph
Linearity	± 1/2% of full scale
Accuracy	±3°
Damping Ratio	0.4 - 0.6
Delay Distance	Less than 3 feet
Temperature Range	-50° C to +85° C

Electrical Characteristics

Input Power	12VDC at 10 mA
Output	0-5 volts for 0-360°
Output Impedance	100 Ohms maximum
Standard Cable Length	300' maximum (consult factory for special cable requirements)

- 2.2 Model 020C Wind Direction Sensor Alignment. (Refer to Figure 2-1 "Typical 020 Installation"). The Orientation screw located on the stem of sensor is to be oriented so that it points south. Proceed as follows:
- A. Tape the sensor hub and column together so that the notches on each part are exactly aligned.
 - B. Loosen the two set screws in the mounting clamp so that the sensor and mounting bushing may be rotated.
 - C. Rotate the sensor so that vane (counterweight end) points directly south.
 - D. A transit located either directly south or directly north will facilitate the alignment process; merely rotate sensor until vane and sensor column merge completely, then tighten the set screws into the mounting bushing.
 - E. Optional alignment accessories are available:
 - 1. A telescopic orientation device, which is inserted into alignment bushing until alignment is achieved. Bushing is locked and orientation device is removed and wind direction sensor installed.
 - 2. A compass orientation device operates in similar manner, but is not suitable for steel towers or poles.
- 2.3 Wiring. The cable assembly contains five wires. Typical wiring hookup is shown in Figure 2-1.
- 2.4 Lightning Protection. Weather sensors are sensitive to direct or nearby lightning strikes. A well-grounded metal rod or frame should be placed above the sensor installation. In addition, the shield on the signal cable leading to the translator must be connected to a good earth ground at the translator end, and the cable route should not be vulnerable to lightning.

3.0 OPERATIONAL CHECK-OUT

3.1 020C Wind Direction Sensor Check-Out (standard 360° operation)

- A. With the sensor connected to the input, upon movement of the sensor's vane in a clockwise direction (as seen from above), the module's output will move up-scale. Movement of the module's output will progress up-scale until the 360° (full scale) mark has been reached; then output will drop to 0°. When the notches on the rotating hub and the column of the sensor are exactly aligned, the output signal will correspond to 180°.

3.2 020C Wind Direction Sensor Check-Out (with 540° conversion electronics)

- A. With the sensor connected to the input, upon movement of the sensor's vane in a clockwise direction (as seen from above), the module's output will move up-scale. Movement of the module's output will progress up-scale until the 540° (full scale) mark has been reached, then the output will drop to 180°. Continued clock-wise rotation will again bring the module's output up-scale to 540° and the action is repeated. When the notches on the rotating hub and the column of the sensor are exactly aligned, the output signal will correspond to 180° (or 540° on the expanded scale).
- B. Check internal heater operation by sliding sensor cover down and touching the housing behind the printed circuit board. The housing should feel warmer than the adjoining metal parts.

- 3.3 If the sensor heater is used, then check internal heater operation by sliding sensor cover down and touching the housing behind the printed circuit board. The housing should feel warmer than the adjoining metal parts. The sensor has a built-in heater that is designed to provide a raise in the internal temperature, providing a small positive pressure. This heater requires an external 12V (@500ma) power supply.

4.2 020C Wind Direction Sensor: 6-12 Month Periodic Service (Refer to Assembly Drawing)

- A. At the crossarm assembly, disconnect the quick disconnect plug from the sensor (leave the cable secured to the crossarm). Remove the sensor from the crossarm assembly bushing. Leave the bushing secure in the fitting.
- B. Loosen the set screw on the bottom edge of the vane hub. Support the rotating hub of the sensor with one hand and pull the vane assembly free.
- C. Visually inspect the wind vane assembly to ensure that the wind vane and counterweight are securely attached to the wind vane rod, and that the set screw in the top edge of the rotating hub holds the rod in place.
- D. Slide the sensor cover down to expose the potentiometer assembly and signal conditioning module.
- E. Inspect the interior of the sensor for any signs of corrosion.
- F. Inspect the potentiometer for loose wires, cracks and corrosion.
- G. Inspect the shaft coupler for proper position. Make sure that the marks on the rotating hub and the column assembly are aligned, and that all special set screws are tight.
- H. Inspect the circuit board assembly for cracks and corrosion. See Section 4.6 B.
- I. Rotate the sensor hub assembly to make sure that it turns freely, and that the sensor bearings are not damaged.
- J. Make sure that the potentiometer assembly is rigid in its holder.
- K. Apply a small amount of silicone lubricant (Dow Corning DC-33 or equivalent) to the sensor O-ring seals. Slide the cover up over the sensor and wipe of any excess lubricant.
- L. Support the rotating hub of the sensor with one hand and mount the wind vane assembly on the hub. The wind vane assembly is keyed to fit the pin on the hub. Tighten the set screw in the bottom edge of the vane hub.
- M. A moisture vent is located on the base of the sensor. Make sure that this vent is clear.
- N. Reinstall sensor according to installation procedure (Section 2.1); verify proper operation using procedures in Section 3.1 and 3.2.

4.5 Sensor Bearing Replacement. (Refer to 020C Sensor Assembly drawing).

A. Remove sensor from tower and remove vane assembly. Refer to Section 4.2.

B. Disassembly sensor and remove old bearings. (#6) (Ref to Assembly Drawing)

1. Slide the sensor cover (18) down to expose the potentiometer assembly (23) and circuit board assembly (15).
2. Make sure that the two (lower) special set screws on the DRIVE COUPLER (21) next to the potentiometer are tight, then loosen the two (upper) special set screws.

NOTE: Any time the shaft coupler assembly is removed, the horizontal potentiometer alignment procedures outlined in Section 4.3 must be performed.

3. Slowly pull the rotating hub/shaft assembly (7) out of the column (8).
4. Remove the shield (4) and slinger (5) from the column.
5. Insert a right-angle type of tool, such as an Allen wrench, into the bearing; cock it slightly to one side and remove the bearing (6).
6. Clean dirt from bearing bores, using a cotton swab and alcohol.

C. Install the new bearing and assemble the sensor.

1. Install new upper bearings in the column assembly. Bearing should slide easily into bearing bore.

CAUTION: Do not lubricate bearing. Do not force.

2. Install slinger (4) and shield (5) on the column (8).
3. Insert the rotating hub/shaft assembly (7) into the column (8), through the shield (4), slinger (5), and lower bearing (6).
4. Support the drive coupler (21) with one hand and slowly push the rotating hub/shaft assembly (7) into it until the shaft bottoms.

D. Recalibrate sensor in accordance with Section 4.3.

E. Replace vane assembly and reinstall (refer to Section 2.1)

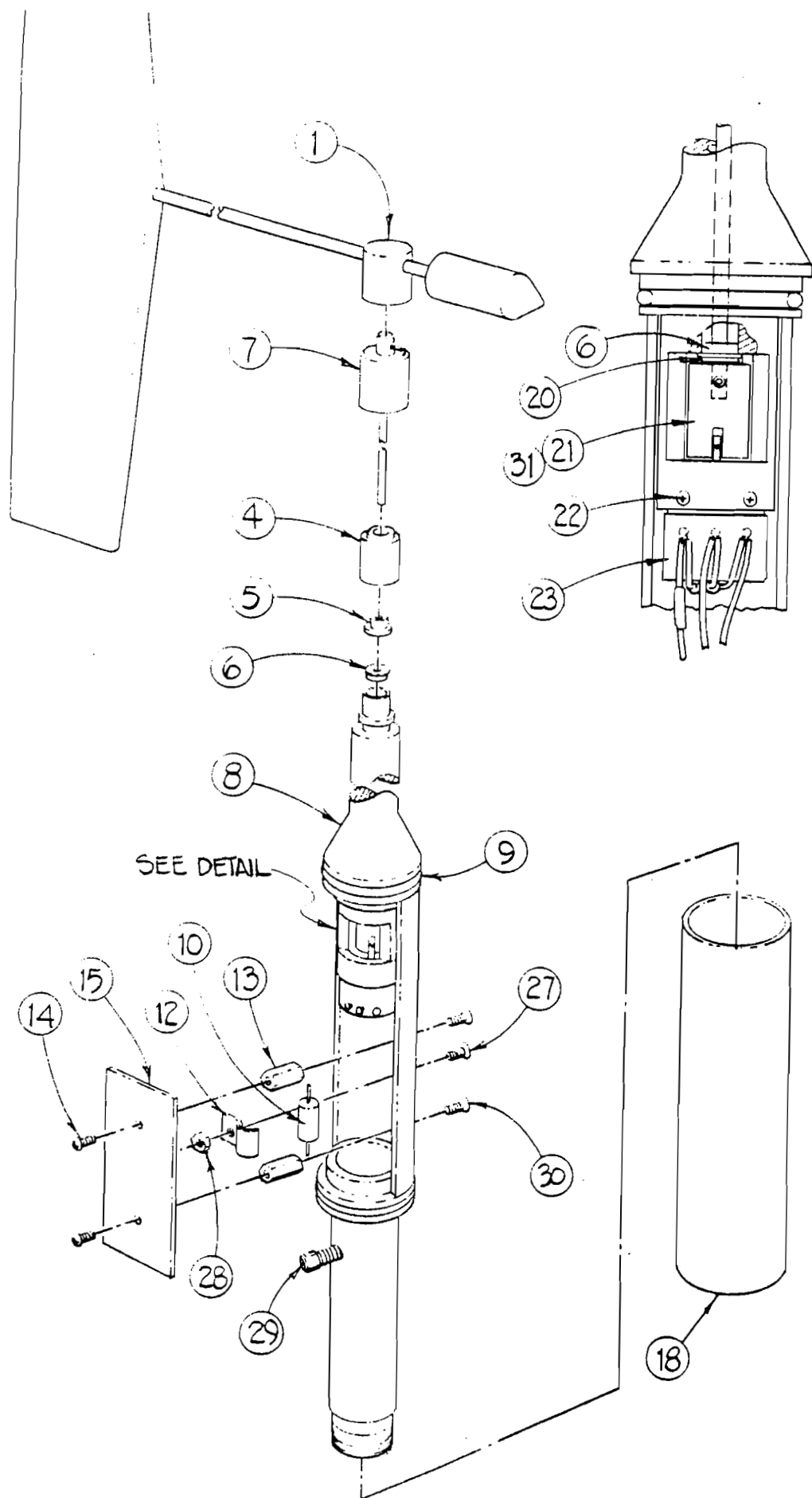
4.8 Integrated Amplifier Replacement

- A. Remove sensor from tower and remove vane assembly (refer to Section 4.2)
- B. Slide the sensor cover (18) down to expose the sensor interior.
- C. Carefully pry amplifier from its socket on the circuit board noting Pin 1 orientation.
- D. Install replacement integrated amplifier by reversing above procedures.
- E. Recalibrate sensor per Section 4.4.

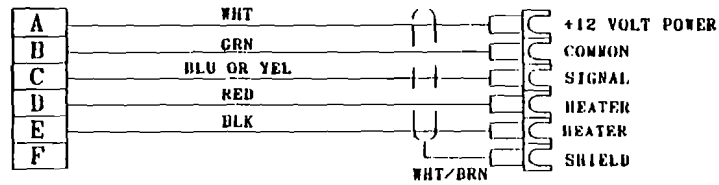
4.9 020C Wind Direction Sensor Repair and Recalibration Service.

The factory provides fast, economical service for the user. This repair and calibration service includes disassembly and detailed inspection of all moving mechanical parts and electronic components.

Service includes replacement of bearings, regardless of apparent condition, calibration, and functional test of sensor. Replacement of the following items, if necessary, is also included: O-rings, shield and slinger, shaft, set screws. Charges for any additional materials will be added to the basic service charge.

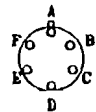
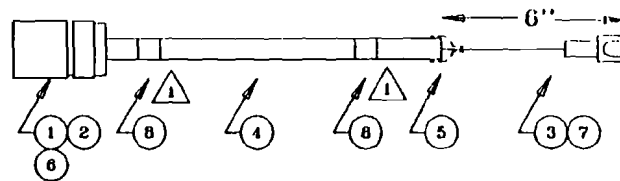


REVISIONS			
REV	DESCRIPTION	DATE	APPROVED
D	REDRAWN PER E.O. 1131	8/11/91	DH
E	WIRE COLORS	10/17/91	DH



SENSOR END

TRANSLATOR END



SOLDER CUP VIEW

ITEM	PART NO.	DESCRIPTION	QTY
1	600351	CONNECTOR, 6 PIN, FEMALE	1
2	480500	CLAMP	1
3	600193	LUG, SPADE, #6	6
4	400014	CABLE, 5 COND., SHIELDED	A/R
5	960050	SLEEVING, 1/4", SHRINK	A/R
6	960075	SLEEVING, 1/8", SHRINK	A/R
7	900610	WIRE, 22 AWG, WHT/BRN	6"
8	960080	SLEEVING, 1/4", CLEAR SHRINK	A/R
9			
10			

△ IDENTIFY CABLE 16" FROM EACH END.
DASH NUMBER = LENGTH IN FEET.

MET ONE INSTRUMENTS			
ASSY, CABLE, 010			
WIND SPEED			
SIZE	FSCH NO.	DWG NO.	REV
		1953	E
SCALE	SHEET 1 OF 1		